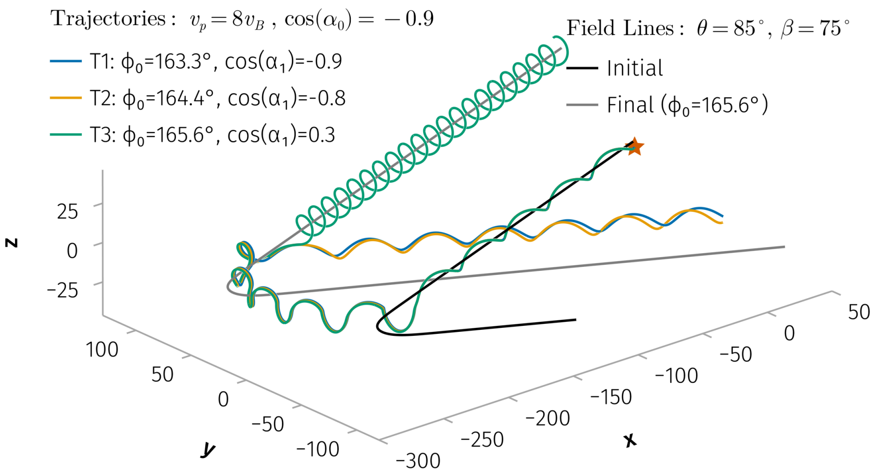
Results

- Interactions with solar wind current sheets drive a distinct regime of energetic particle transport.
- Current sheets induce non-adiabatic scattering and rapid cross-field diffusion



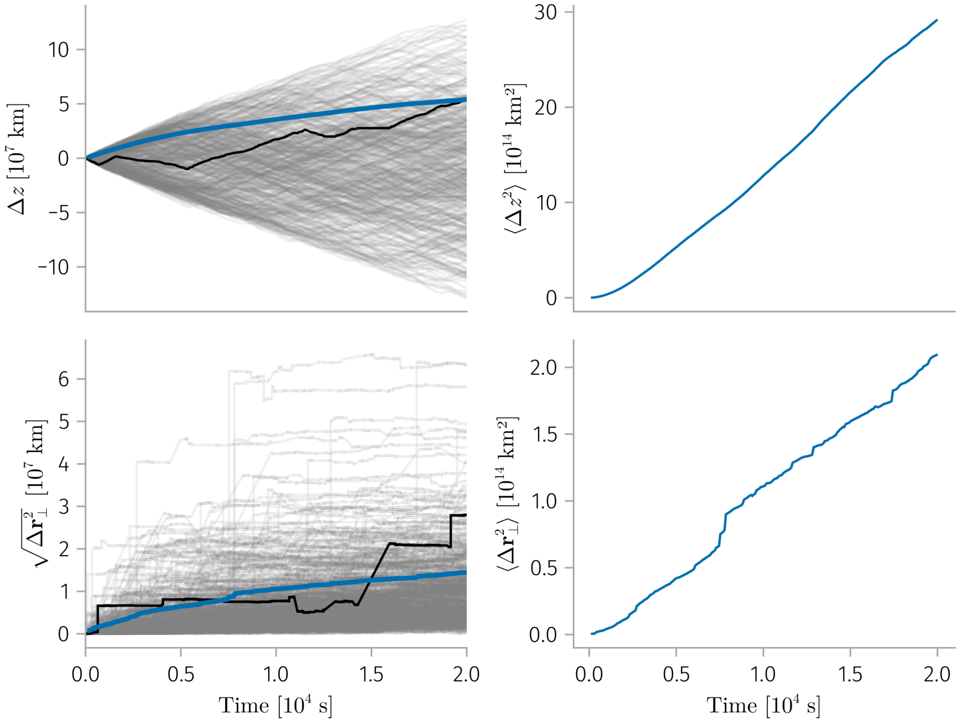


Figure 2: Parallel and perpendicular displacements for 1-MeV particles interacting with near-Earth current sheets. Gray lines: individual particle trajectories; Blue line: standard deviation of the ensemble; Black line: a representative trajectory.

Results

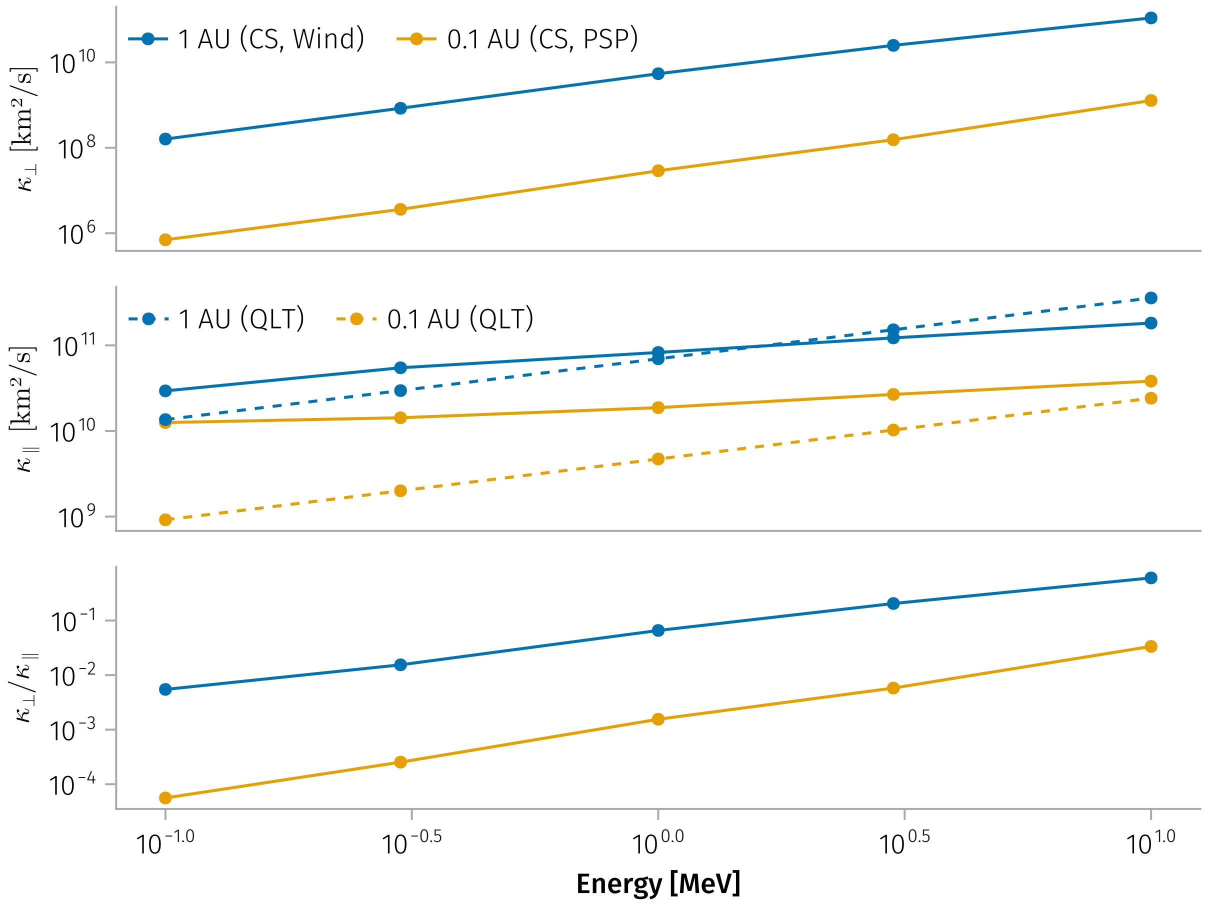


Figure 3: Diffusion coefficients vs. particle energy at 1 AU and 0.1 AU. (a) Parallel diffusion; (b) Perpendicular diffusion; (c) Ratio of perpendicular to parallel diffusion.